

## Automating Infrastructure with Terraform

This task is designed to **showcase** your IaC and automation proficiency. The ultimate goal is to develop Terraform code capable of provisioning applications on Kubernetes.

Strive for code reusability when performing the task. Leverage Terraform's features to minimize code duplication.

### Task Requirements

Utilizing Terraform setup a local Kubernetes cluster.

- Deploy **N** basic web applications
  - The application should serve web requests and return the following
    - Pod name
    - Pod IP address

#### Networking Requirements:

- Expose a local endpoint that dynamically routes traffic to the different applications
- Traffic should be routed to pods capable of responding to requests
- Ensure each application has a distinct and accessible route

**You can choose any tool/application you want that answers the requirements.**

### Bonus tasks

Implement an additional application stack with minimal additional lines of code.

- Deploy a second application
- Deploy a third application such as [podinfo](#)

#### CI/CD:

- Create CI/CD GitHub Action for the above (This can run locally using Act or another method of your choice)

## Submission Guidelines

Submit your work via a public GitHub repository. Ensure your repository includes clear documentation and instructions for setting up and executing the code.

Additionally, provide any necessary configuration files and **README.md** documentation detailing the purpose, usage, and deployment steps for your project.

- If you used AI, explain **how** you used it and what prompts or guidance you gave it
- Include notes and comments if it will help us understand your design



## Time Estimate

This assignment is expected to take approximately 2–5 hours.

Don't hesitate to reach out if you have any questions or need further clarification.